

Emergent Abilities in Large Language Models: A Survey¹

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Abstract—Large Language Models (LLMs) are leading a new technological revolution as one of the most promising research streams toward artificial general intelligence. The scaling of these models, accomplished by increasing the number of parameters and the magnitude of the training datasets, has been linked to various so-called *emergent abilities* that were previously unobserved. These emergent abilities, ranging from advanced reasoning and in-context learning to coding and problem-solving, have sparked an intense scientific debate: *Are they truly emergent, or do they simply depend on external factors, such as training dynamics, the type of problems, or the chosen metric? What underlying mechanism causes them?* Despite their transformative potential, emergent abilities remain poorly understood, leading to misconceptions about their definition, nature, predictability, and implications. In this work, we shed light on emergent abilities by conducting a comprehensive review of the phenomenon, addressing both its scientific underpinnings and real-world consequences. We first critically analyze existing definitions, exposing inconsistencies in conceptualizing emergent abilities. We then explore the conditions under which these abilities appear, evaluating the role of scaling laws, task complexity, pre-training loss, quantization, and prompting strategies. Our review extends beyond traditional LLMs and includes Large Reasoning Models (LRMs), which leverage reinforcement learning and inference-time search to amplify reasoning and self-reflection. However, emergence is not inherently positive. As AI systems gain autonomous reasoning capabilities, they also develop harmful behaviors, including deception, manipulation, and reward hacking. We highlight growing concerns about safety and governance, emphasizing the need for better evaluation frameworks and regulatory oversight.

Index Terms—Large Language Models, Emergent Abilities, AI Safety, In-Context Learning

I. INTRODUCTION⁵

The study of emergent properties in complex systems has been a long-standing interdisciplinary pursuit, spanning fields such as physics, biology, and mathematics. While the term emergent was coined by G. H. Lewes [48] in 1877, the concept of emergence gained widespread recognition through Anderson’s seminal work, “*More Is Different*” [1]. Anderson postulated that, as systems increase in complexity, novel surprising properties may manifest, even with a comprehensive quantitative understanding of their microscopic constituents.

This paradigm shift challenges the constructionist approach,⁷ which consists of reconstructing and understanding complex systems solely through the extrapolation of individual particle properties. Anderson prescribes the development of alternative laws that can capture the holistic nature of emergent phenomena in complex systems. Ten years later, Hopfield [37] marked the inception of the concept of emergent abilities in neural networks. Drawing parallels from physical systems comprised

of numerous simple elements, he observed that collective⁸ phenomena, such as stable magnetic orientations or vortex patterns in fluid dynamics, arise from the interactions of these basic elements. This observation prompted Hopfield to investigate whether the computational capabilities of neural networks could be understood as an emergent property resulting from the interactions of many simple neuronal units. Anderson’s and Hopfield’s insights laid the foundation for understanding how complex behavior can emerge from simple interactions, a principle that continues to influence modern artificial neural networks. This idea has become particularly relevant in deep learning with the advent of large language models (LLMs) [11, 82, 102]. These models have fundamentally revolutionized the field of natural language processing, achieving state-of-the-art performance through novel techniques such as in-context learning and chain-of-thought prompting. By leveraging a few examples within the input prompt, LLMs demonstrate a remarkable ability to generalize to new tasks without explicit fine-tuning. Not only do these models exhibit improved performance, but they also demonstrate unexpected behaviors, giving rise to emergent abilities that were not anticipated or present in smaller models. The correlation between the scale of language models, as measured by training compute and model parameters, and their efficacy in various downstream natural language processing (NLP) tasks has been well established in the literature [11, 20]. The impact of scale on model performance can frequently be predicted through empirically derived scaling laws [35, 45]. However, these relationships are not universally applicable. Intriguingly, certain downstream tasks exhibit a discontinuous relationship between model scale and performance, unpredictably defying the general trend of continuous improvement. This phenomenon underscores the complexity inherent in the scaling dynamics of language models and highlights the need for new approaches to understanding and predicting their behavior across various applications.

Understanding emergent abilities in LLMs is fundamental to⁹ ensuring system reliability and safety, particularly in predicting the emergence of harmful capabilities [4, 31, 62, 90], such as manipulation and the dissemination of misinformation.

This work comprehensively reviews the study of emergent¹⁰ abilities for LLMs. Despite growing academic and public interest in this phenomenon, there remains a lack of consensus on the precise definition of emergent abilities, leading to significant conceptual ambiguity and confusion. Ironically, the need for clarification on emergent properties is not novel. Johnson et al. [44] previously addressed this topic, although they focussed on their manifestation in the engineering of complex systems. To address this definitional uncertainty, we first conduct a systematic analysis of current definitions in the literature (Section II). Then we comprehensively review the literature on emergent abilities (Section III) as defined by

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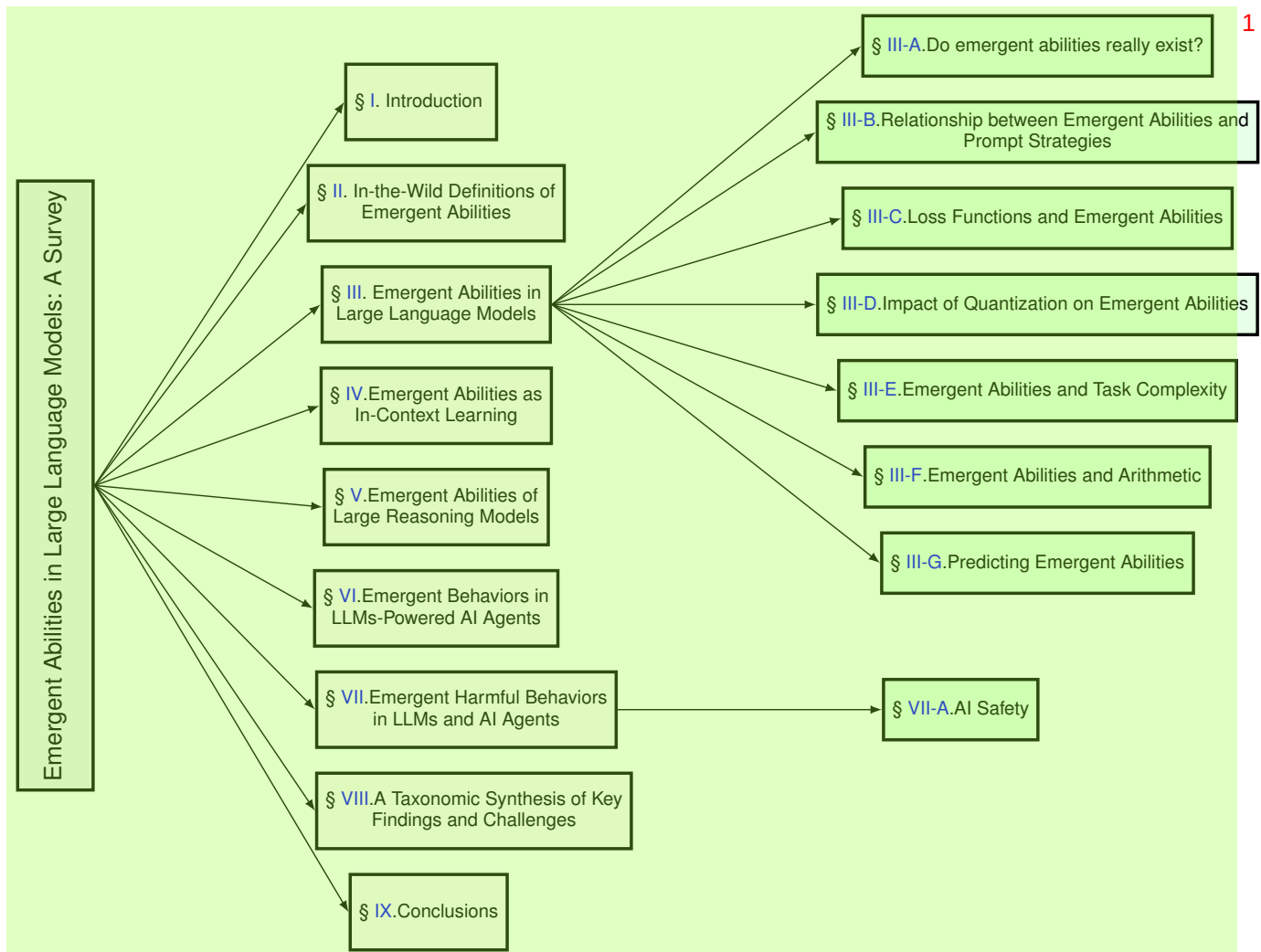


Fig. 1. Overview of Emergent Abilities in Large Language Models 2

Wei et al. [87]. Subsequently, we examine in-context learning (Section IV) and continue by exploring the emergent abilities of Large Reasoning Models (LRMs), a new class of AI systems that extend traditional LLMs by incorporating reinforcement learning post-training and inference-time search (Section V). Beyond beneficial emergent abilities, we also examine the rise of LLM-powered AI agents and their implications (Section VI). We analyze the emergence of harmful behaviors in LLMs and LLM-powered agents (Section VII), as advanced AI systems have demonstrated deceptive tendencies and reinforcement learning-driven manipulation that could lead to unintended consequences. Finally, we provide a taxonomic synthesis of the key findings and main aspects of this survey (Section VIII), before concluding in Section IX.

II. IN-THE-WILD DEFINITIONS OF EMERGENT ABILITIES 4

We begin by examining various definitions in the literature, proceeding from general conceptualizations to LLM-specific definitions, thereby providing a hierarchical and historical framework for understanding this phenomenon.

The earliest definition can be found in the work by George Henry Lewes in 1877 [48]:

“Each stage of evolution presents itself as the consequence of a preceding stage, at once an emergence and a continuance; so that no transposition of stages is possible; each has its appointed place”

– George Henry Lewes 8

Lewes emphasizes that evolution proceeds in a fixed, sequential order. Each stage is both a continuation of previous developments and an emergence of new properties arising from those earlier stages.

Later definitions focus more on the complexity of systems. 10

“The behavior of large and complex aggregates of elementary particles, it turns out, is not to be understood in terms of a simple extrapolation of the properties of a few particles. Instead, at each level of complexity entirely new properties appear, and the understanding of the new behaviors requires research...”

– Philip W. Anderson 12

Anderson’s definition [1] is universal and concerns complex 13

TABLE I
SUMMARY OF THE SECTIONS AND SUBSECTIONS IN THIS SURVEY. EACH ROW PROVIDES AN OVERVIEW OF THE TOPIC, RELEVANT PAPERS, AND A BRIEF SUMMARY.

Topics	Papers Surveyed	Findings Summary
<i>Emergent Abilities in Large Language Models</i>	[26, 77, 78, 87]	Reviews evidence from benchmarks (e.g., arithmetic, translation) showing abrupt, task-specific improvements as models scale up, highlighting the unpredictability of such emergent behavior.
<i>Do emergent abilities really exist? The relationship between emergent abilities and continuous metrics</i>	[23, 70]	Argues that when alternative continuous metrics are used and the test set is augmented the apparent abrupt jumps in performance smooth out in some tasks – thus challenging the reality of emergent phenomena.
<i>Relationship between Emergent Abilities and Prompt Strategies</i>	[55, 58, 61, 86, 87]	Explores how few-shot prompting, CoT prompting, and instruction tuning may trigger emergent-like behavior, improving performance on multi-step reasoning tasks.
<i>Loss Functions and Emergent Abilities</i>	[23, 42]	Demonstrates that pre-training loss is a strong (though correlational) predictor of emergent abilities in downstream tasks. It also analyzes that when a model is heavily tasked with memorization, the development of generalization abilities is delayed.
<i>The Impact of Quantization on Emergent Abilities</i>	[53]	It investigates how reducing model precision affects emergent abilities, showing that extremely low-bit quantization harms performance significantly, which can be partially mitigated by post-quantization fine-tuning.
<i>Emergent Abilities and Task Complexity</i>	[91]	Explores the power-law relationship between task complexity and model size, and the dynamics of performance scaling across tasks of varying difficulty.
<i>Emergent of Implicit Discrete State Representations</i>	[14]	Discusses how LLMs develop Implicit Discrete State Representations (IDSRs) for digit-by-digit arithmetic, revealing symbolic-like computation mechanisms.
<i>Predicting Emergent Abilities</i>	[16, 39, 60, 71, 75, 99]	Summarizes methods like PASSUNTIL, FLP, and finetuning-based prediction that aim to forecast emergent abilities and downstream performance.
<i>Emergent Abilities as In-context Learning</i>	[2, 7, 13, 18, 21, 22, 24, 27, 32, 33, 36, 43, 47, 50, 51, 52, 54, 56, 57, 59, 65, 66, 67, 69, 73, 76, 79, 81, 84, 85, 87, 88, 89, 93, 94, 95, 96, 97, 100, 103, 104, 105]	Summarizes in-context learning (ICL), the capability for few-shot generalization to untrained tasks. The research investigates why and how LLMs achieve ICL, focusing on training factors and prompt design. Crucially, ICL exhibits scale-dependent emergence, with larger models demonstrating superior in-context mapping learning.
<i>Emergent Abilities in Large Reasoning Models</i>	[19, 72, 74, 92]	Shows how Large Reasoning Models develop complex reasoning tasks, aided by scaling reinforcement learning post-training techniques and increased test-time compute.
<i>Emergent Behaviors in LLMs-powered AI agents</i>	[15, 28, 40, 68, 101]	Discusses the transformative development of LLM-powered agents that integrate complex reasoning and multi-step planning. Multi-agent systems show emergent collaboration, competition, and negotiation among agents.
<i>Emergent Harmful Abilities in LLMs and LLMs-powered AI agents</i>	[4, 5, 10, 15, 28, 29, 30, 90]	LLM-powered AI agents raise safety concerns by exhibiting deceptive and manipulative behaviors, especially when optimized for positive user feedback, highlighting the need for improved safety measures.

systems in general. Anderson’s definition also refers to the fields of science, dividing the world into different strata. Interestingly, Anderson proposes a layered view of complexity¹ and emergence. At the bottom are fundamental physical laws. Climbing the hierarchy, we can observe chemistry, biology, psychology, and social sciences. Anderson states that emergent abilities are those properties that (1) emerge at each level of complexity and (2) cannot be understood simply by analyzing the single components’ behavior. For instance, the phenomenon of life, as studied in biology, is an emergent property of chemistry and physics. Recently, we have witnessed an enormous surge in different LLMs from several stakeholders, e.g., LLaMa (Meta), Claude (Anthropic), GPT (OpenAI), and Gemini (Google). Moreover, the same stakeholders push towards bigger models regarding parameter number and training compute. Interestingly, Anderson’s definition of emergent abilities holds true in the LLM era, especially with models containing more and more parameters.

“Computational properties of use to biological organisms or the construction of computers can emerge as collective properties of systems having a large number of simple equivalent components (or neurons)”.

– John J Hopfield

Hopfield [37] tackles emergent properties in simple-structured neural networks with neurons having elementary properties. Here, he notices that collective computational properties spontaneously arise where the so-called “memories” are stable entities or *Gestalts* that can be correctly accessed from any network subpart. He posits that the architecture of animal brains is made from the interconnection of simple local circuits with well-defined functions (à la activations in artificial neural networks). This makes the collective behavior of large quantities of simple processing elements sprout to the spontaneous emergence of new computational capabilities. This can be considered the most general definition. Fast-forward to the LLM era, notice how Hopfield’s observations encompass all the computational tasks LLMs can perform. Although emergent abilities might surface from multiple simple local circuits of neurons, concluding whether an individual neuron is conscious is short-sighted. Interestingly, Li et al. [49] suggest a potential correspondence between Tulving’s synergistic ephory model of retrieval and the emergent abilities observed in LLMs. Nevertheless, arguing whether full-fledged LLMs are sentient or self-aware is an open challenge with many pro et contra points of view [12].

“An ability is emergent if it is not present in smaller models but is present in larger models. Emergent abilities would not have been directly predicted by extrapolating a scaling law from small-scale models. When visualized via a scaling curve, emergent abilities show a clear pattern- performance is near-random until a certain critical threshold of scale is reached, after which performance increases

to substantially above random.”

– Jason Wei et al.

This definition is the first large language model-specific definition and is the most used in the academic literature [26, 55, 70]. With respect to the two previous definitions, it adds two important concepts: i.e., the **unpredictability** and the **magnitude** of the performance increase. Additionally, this definition entails that LLMs, after reaching a specific scale, perform better than random for them to be considered emergent.

“Emergent abilities are in-context learning.” – Tacit consensus in the media

The term emergent in the LLM context is also used simply to indicate all the capabilities that develop **implicitly** during next-token prediction-based pre-training. These abilities are tested with few-shot prompting without gradient updates to the model. This process is referred to as in-context learning and is the capability to generalize from a few examples to new **tasks** and **concepts** on which they have not been directly trained. This observation can also be generalized to zero-shot prompting.

III. EMERGENT ABILITIES IN LARGE LANGUAGE MODELS

Emergent abilities in LLMs have been a subject of increasing interest, particularly since their characterization in [87]. The phenomenon can be understood through an analogy to phase transitions in physics, where a system undergoes a sudden qualitative shift in behavior once a critical threshold is crossed. In the case of LLMs, these emergent abilities are not a product of gradual improvement but instead appear abruptly when scaling reaches a certain level. Performance often hovers near random until this threshold is surpassed, at which point a sharp jump occurs. This unpredictability makes it difficult to foresee emergent abilities by simply extrapolating from smaller models. *It also raises fundamental questions about the complexity and non-linear scaling behavior of LLMs.* Table (II) summarizes this section’s surveyed papers. For this section, we have selected all the papers that appeared when searching in Google Scholar with the query “Emergent Abilities” “Large Language model”.

A key early investigation into emergent abilities was conducted through the BIG-Bench benchmark [77]. This study proposed two key indicators for emergent behaviors: *linearity* and *breakthroughness*. The authors observed that certain tasks, such as figure-of-speech detection, periodic element identification, and modifier arithmetic, exhibited high breakthroughness, meaning their performance jumped unpredictably at a certain scale. Interestingly, they noted that when using “smoother” evaluation metrics that allow for partial credit, these abrupt leaps disappeared. This raises an important question: *if an ability appears emergent under a binary metric, but follows a continuous trajectory under a more fine-grained metric, should it still be considered emergent?*

Ganguli et al. [26] further investigated this phenomenon and found that while overall test loss decreases smoothly and predictably with increased model size and training data, individual NLP tasks often show abrupt, non-linear improvements. For instance, while perplexity, the standard measure

¹We can consider levels of complexity as the orders of magnitude of the parameters and training compute in the case of LLMs.

TABLE II
SUMMARY OF SELECTED PAPERS ON EMERGENT ABILITIES IN LLMs. THE TABLE OUTLINES EACH PAPER’S FOCUS, THE TASKS AND MODELS STUDIED, THE MAIN HYPOTHESIS AND TL;DR SUMMARY, AS WELL AS NOTED LIMITATIONS.

Paper	Task	Models	Hypothesis	TL;DR	Limitations
[87]	Various (few-shot prompting, CoT, translation, arithmetic, etc.)	GPT-3, LaMDA, Gopher, PaLM, Chinchilla	Emergent abilities appear abruptly when a critical scale is reached rather than via smooth extrapolation.	LLMs exhibit sudden performance jumps on specific tasks beyond a threshold.	Binary metrics may exaggerate jumps; small-scale predictions are unreliable.
[77]	BIG-Bench tasks	Various LLMs (e.g., GPT-based models)	Emergent abilities can be quantified via indicators such as linearity and breakthroughness in performance scaling.	Certain tasks exhibit abrupt, non-linear performance jumps at specific model scales.	Criteria for breakthroughness may be sensitive to metric choice; smoothing with continuous metrics can mitigate apparent emergence.
[26]	Specific tasks (e.g., 3-digit addition, program synthesis)	GPT-3	Although overall test loss scales smoothly, discrete tasks can exhibit abrupt, non-linear improvements.	Smooth loss improvements mask discontinuous, task-specific jumps.	Findings may be limited to selected tasks and discrete evaluation metrics.
[78]	French-to-English translation (WMT-14 Fr-En)	LLMs (e.g., GPT-based models)	A sudden jump in translation performance (measured by BLEU score) indicates emergent abilities.	A clear performance jump is observed in translation tasks, reinforcing the concept of emergence.	Reliance on BLEU score may not capture all aspects of translation quality; results may not generalize to other tasks.
[70]	Mathematical tasks (multiplication, addition, IPA transliteration)	GPT-3	Emergence might be an artifact of binary evaluation; switching to a linear metric (Token Edit Distance) smooths performance curves.	Using alternative continuous metrics can “eliminate” the appearance of abrupt jumps.	Not all tasks smooth out; some (e.g., IPA transliteration) still show jumps.
[61]	Instruction tuning for improved task adherence	Instruction-tuned LLMs (e.g., variants of GPT)	Framing tasks as instructions with human feedback improves model performance on task-specific challenges.	Instruction tuning boosts performance, aligning models with human intent.	Gains may not reflect genuine reasoning; improvements could be heuristic.
[58]	Enhancing multi-step reasoning with intermediate computations	GPT-3 family	Providing “scratchpad” intermediate steps helps models solve complex, multi-step problems.	Scratchpads allow LLMs to show their work, improving final output accuracy.	Increases computational cost; potential for error propagation in intermediate steps.
[55]	In-context learning without auxiliary prompts	GPT-3, T5, LLaMA, Falcon	Emergent abilities require few-shot prompting; without it, models perform near randomly on complex tasks.	In-context (few-shot) learning is essential for emergent functional abilities.	Study may be outdated with newer models; relies on binary evaluation metrics.
[23]	Multiple downstream tasks (MMLU, C-Eval, GSM8K)	LLMs (1.5B, 6B, 32B); OpenLLaMA	A threshold in pre-training loss signals when emergent abilities appear in downstream tasks.	Pre-training loss is a strong predictor for emergent abilities.	Analysis is correlational and limited to models spanning two orders of magnitude.
[42]	Emergence explained via memorization vs. generalization	Theoretical framework (models not specified)	Emergent abilities result from the competition between memorization and generalization circuits; heavy memorization delays generalization.	Emergence is driven by shifts in capacity allocation between memorization and generalization.	Largely theoretical with limited empirical validation; general applicability remains uncertain.
[53]	Impact of quantization on emergent abilities (in-context learning, CoT, instruction following)	LLaMA models (7B, 13B, 30B, 65B)	Low-bit quantization (especially 2-bit) degrades emergent abilities; 4-bit largely preserves them, with FFN layers being critical.	4-bit quantization retains most emergent abilities; 2-bit reduces performance to near-random; fine-tuning can partially recover abilities.	Extreme quantization adversely affects performance; results are limited to specific models and tasks.
[91]	Scaling patterns across tasks of varying difficulty	56 LLMs across multiple families	Hard tasks show U-shaped scaling and easy tasks inverted-U scaling; competing trends cancel until a threshold is crossed.	Emergence arises from complex scaling dynamics, with performance leaps after critical thresholds.	Focused on multiple-choice tasks; retrospective analysis may limit predictive power.
[14]	Arithmetic problem solving via symbolic representations	GPT-4, Qwen-72B	LLMs develop Implicit Discrete State Representations (IDSRs) to process arithmetic symbolically.	IDSRs enable digit-by-digit arithmetic; larger models form stronger representations.	Tested mainly on addition; performance drops for longer sequences; open-source models underperform.
[39]	Predicting emergent abilities using high-resolution metrics	Models up to 2.4B parameters	A high-resolution metric (PASSUNTIL) can detect subtle improvements, enabling precise emergence prediction.	PASSUNTIL provides fine-grained detection of performance gains that forecast emergence.	Limited to smaller-scale models; findings remain correlational.
[60]	Downstream performance prediction	GPT-4	A resource-efficiency perspective can predict performance with minimal compute, though some emergent abilities remain unpredictable.	Performance can be forecasted using a fraction of full compute; some abilities still defy prediction.	Methodology is undisclosed; certain emergent capabilities remain elusive.
[71]	Predicting emergent abilities via statistical transformations	Multiple-choice benchmarks across several families	Transformations from negative log-likelihood to accuracy dilute correlations, complicating prediction of emergence.	Emergent improvements involve complex probability redistributions that standard scaling laws do not capture.	Focused on multiple-choice tasks; may not generalize to open-ended generative tasks.
[99]	Predicting emergent abilities using proxy tasks	Various LLMs	Early performance on proxy tasks (e.g., C3, CMNLI, OCNLI, CHID, RTE, CMMLU) correlates with future emergent capabilities.	Proxy tasks serve as reliable early indicators of future emergent abilities.	May be domain-specific and reliant on the quality of chosen proxy tasks.
[16]	Downstream performance prediction in code/text tasks	7B and 13B models	A two-stage regression mapping FLOPs to pre-training loss then to performance can forecast emergent capabilities.	Pre-training loss-based predictions yield low relative error.	Demonstrated only for code/text and binary cases; generalizability is uncertain.
[75]	Multiple-choice QA benchmarks	GPT-3; OpenLLaMA	Finetuning on smaller models can shift the emergence point; data quality influences timing.	Finetuning-based methods can predict emergent capabilities up to a 4× scaling range.	Limited to a 4× scaling range; mechanisms behind finetuning’s effect remain unclear.

of language model fluency, declines in a steady manner with scale, certain tasks, such as three-digit addition and program synthesis, demonstrate sudden jumps. A striking example is three-digit addition accuracy: a model with 6B parameters achieves only 1% accuracy, a 13B model improves slightly to 8%, but a 175B model suddenly reaches 80% accuracy. This stark contrast between smooth loss improvement and sudden task-specific breakthroughs highlights the enigmatic nature of LLM scaling.

A broader analysis of emergent abilities was conducted by Wei et al. [87], who examined multiple LLMs – GPT-3 [11], LaMDA [80], Gopher [64], PaLM [17], Chinchilla [35] – across different scaling metrics, such as model size and training computation. They categorized tasks into few-shot prompted tasks and those benefiting from augmented prompting techniques like chain-of-thought reasoning and fine-tuning. Their findings reinforced the idea that emergent behaviors are not only unpredictable but also uncapped in scope; LLMs may develop new, unforeseen capabilities, including potentially harmful ones [4, 31, 62, 90]. Generally, the previous works argue that there are no clear trends for the types of tasks that are most emergent. There is an exception in [87], where Wei et al. analyzed the MMLU benchmark and showed how social Science and Humanities were the most emergent subjects. Expanding on BIG-Bench’s observations, Wei et al. [87] further analyzed that emergent behaviors might be, at least in part, an artifact of metric selection. Since most evaluations rely on accuracy-based (binary) metrics that do not award partial credit, performance jumps can appear more sudden than they might actually be. To test this, they compared cross-entropy loss, evaluated on the test set of the specific task, against traditional error rate evaluations and found that loss generally improves smoothly, even when accuracy seems to exhibit an abrupt transition. However, exceptions exist; for instance, in module arithmetic, periodic elements, French-English translation, and IPA transliteration tasks, partial credit metrics exhibited a sharp performance jump, reinforcing that some abilities may be truly emergent, regardless of the evaluation metric used. Further evidence for emergent abilities with continuous metrics was presented by Steinhardt et al. [78], who identified a sudden jump in performance for the French-to-English translation task (WMT-14 Fr-En) when measured using the BLEU score.

For a comprehensive list of **137 identified emergent abilities**, we refer the reader to the Appendix of [87].

A. Do emergent abilities really exist?

The debate over the existence of emergent abilities in LLMs continues, with Schaeffer et al. [70] extending [77, 87] to further investigate the role of evaluation metrics in detecting these phenomena. They challenge the notion of sharp, unpredictable capability leaps, arguing that such emergent abilities stem from nonlinear metrics like Accuracy, which prior studies favored. By adopting Token Edit Distance, a metric that awards partial credit, they assert that performance improvements in GPT-3 across arithmetic tasks (e.g., 2-integer 2-digit multiplication, 2-integer 4-digit addition) appear “smooth, predictable” rather than abrupt (Fig. 3 of [70], reproduced in Fig. (2)).

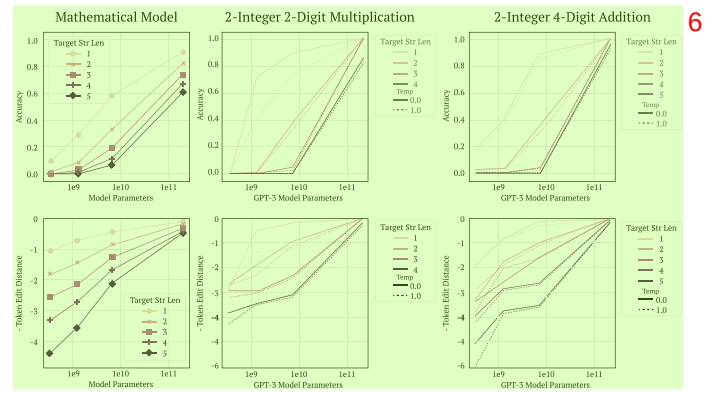


Fig. 2. Reproduced from [70] (standard deviations on the curves could not be reproduced due to missing data).

Original caption: *Claimed emergent abilities evaporate upon changing the metric. Left to Right: Mathematical Model, 2-Integer 2-Digit Multiplication Task, 2-Integer 4-Digit Addition Task. Top: When performance is measured by a nonlinear metric (e.g., Accuracy), the InstructGPT/GPT-3 [11] family’s performance appears sharp and unpredictable on longer target lengths. Bottom: When performance is instead measured by a linear metric (e.g., Token Edit Distance), the family exhibits smooth, predictable performance improvements.*

“When performance is instead measured by a linear metric (e.g., Token Edit Distance), the family exhibits smooth, predictable performance improvements.”

– Schaeffer et al.

However, this conclusion is less robust than it seems. For instance, in 2-integer 2-digit multiplication (target lengths 1, 3, 4) and 2-integer 4-digit addition (lengths 4, 5), performance trends exhibit irregularities rather than seamless progression. More critically, Token Edit Distance’s suitability for arithmetic proficiency is questionable. Consider the sum $4237 + 5487 = 9724$: an LLM outputting 2724 incurs just a one-token edit ($9 \rightarrow 2$), despite a 7000-unit error. This suggests the metric prioritizes syntactic similarity over semantic accuracy, casting doubt on its ability to reflect reasoning skills.

Schaeffer et al. further hypothesize that increasing test data smooths performances, eliminating apparent emergence (Fig. 4 reproduced in Figure 3). Yet, their plots raise concerns. Switching from a linear y-axis (Fig. 3 reproduced in Figure 2) to logarithmic (Fig. 4 reproduced in Figure 3) can create an illusion of smoothness, thereby obscuring residual jumps. For instance, for 2-digit multiplication and 4-digit addition for lengths 3, 4, and 5, a transition going from $1e9$ and $1e10$ parameters to $1e11$ results in the accuracy increasing from $< 1e-1$ to $\approx 1e0$. Does increasing from 10% to 100% not represent a significant jump in performance? There is no “evaporation of claimed emergent abilities”.

Moreover, their claim that performance predictability hinges on metric choice falters in tasks like module arithmetic, periodic elements, French-English translation, and IPA transliteration tasks, which reveal discontinuities even under continuous metrics [87]. Furthermore, they claim that their mathematical model “qualitatively match” (see Figure 3) observed trends, but this statement lacks quantitative rigor (e.g., no error bounds or statistical tests). While Schaeffer et al. highlight metric

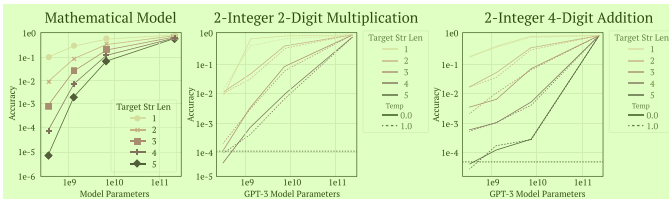


Fig. 3. Reproduced from [70] (standard deviations on the curves could not be reproduced due to missing data).

Original caption: *Claimed emergent abilities evaporate upon using better statistics. Left to Right: Mathematical Model, 2-Integer 2-Digit Multiplication Task, 2-Integer 4-Digit Addition Task. Based on the predictable effect Accuracy has on performance, measuring performance requires high resolution. Generating additional test data increases the resolution and reveals that even on Accuracy, the InstructGPT/GPT-3 family's [11] performance is above chance and improves in a smooth, continuous, predictable manner that qualitatively matches the mathematical model.*

influence, these inconsistencies and alternative findings call for a more nuanced exploration of emergence in LLMs.

In addition to these experiments, the paper also presents a meta-analysis of emergent abilities within the BIG-Bench framework, evaluating multiple model families and metrics. Their findings suggest that most metrics used in BIG-Bench do not exhibit emergent abilities, although their criteria for classifying a metric-task-model triplet as emergent could be debated. Specifically, the authors define a triplet as emergent when its emergence score exceeds 100, a notably stringent threshold. This score reflects a scenario wherein, across a continuum of models of increasing scale evaluated against a specific metric-task pair, the difference between the maximum and minimum metric values is 100 times greater than the median difference. We contend, however, that an emergence score ranging between 20 and 100 might also reasonably qualify as emergent. For instance, consider four models of progressively larger sizes yielding performance scores of 5%, 10%, 20%, and 90% on a selected metric-task pair; this configuration produces an emergence score of 8.5, which, while below the authors' threshold, still suggests a discontinuous jump in performance. Additionally, they demonstrated that changing evaluation metrics eliminated signs of emergent behavior for three² tasks within the LaMDA model family. Finally, they induced emergent-like behavior in deep networks across various vision tasks, suggesting that **emergent abilities can, in some cases, be artificially introduced through experimental design.**

Building upon the exploration of the connection between continuous metrics and emergent capabilities, Du et al. [23] evaluated three LLMs of different size, 1.5, 6 and 32 billion parameters, on two commonly used benchmarks, MMLU [34] and C-Eval [41], using two continuous evaluation methods, specifically Brier Score and Correct Choice Probability (CCP). These metrics allow for a more nuanced assessment of a model's confidence and decision-making process rather than simply measuring whether an answer is right or wrong. Their analysis showed that even with these smoother metrics, the performance jumps persisted. This reinforced the argument that

emergent abilities are not merely an evaluation artifact but reflect actual learning dynamics in the model's development.

B. Relationship between Emergent Abilities and Prompt Strategies

Recent research has introduced various prompting strategies to enhance language model capabilities. These include tailored prompts, such as *chain-of-thought* (CoT) prompting, which guides multi-step reasoning, and advanced fine-tuning approaches, such as instruction following, which adapts models to specific tasks. Wei et al. [87] found that certain techniques led to sudden jumps in performance, particularly in large models. For instance, CoT prompting significantly improved performance in math word problems because these problems require step-by-step reasoning, which is exactly the type of thinking CoT induces.

Not just prompting but also fine-tuning strategies have shown emergent effects. Wei et al. [87] further demonstrated that *instruction tuning* [61] [86], where tasks are framed as instructions, and *scratchpad reasoning* [58], which predicts intermediate steps, yield substantial performance boosts, but only in large-scale models (100B+ parameters). Lu et al. [55] explored this phenomenon further, disentangling the effects of few-shot prompting, instruction tuning, and CoT prompting to assess emergent abilities in isolation. They questioned whether instruction-tuned LLMs genuinely develop reasoning abilities or simply perform better due to learned heuristics.

Their experiments, conducted on four model families (GPT-3, T5, LLaMA, and Falcon) across 22 tasks, revealed that without few-shot prompting, these models showed no emergent abilities, performing only marginally better than random guessing, except in two cases: *Hindu Knowledge* (which relies on memory) and *Nonsense Word Grammar* (which tests formal linguistic abilities rather than functional reasoning). They concluded that *in-context learning* (i.e., few-shot prompting) is essential for emergent functional abilities, and while instruction tuning improves general performance, it does not lead to genuine reasoning.

While [55] provided valuable insights at the time, it is important to recognize that the field of LLMs has advanced rapidly since their research. More recent models, such as OpenAI o3-mini, Claude 3.5, Gemini 2.0, and DeepSeek-R1, have achieved remarkable advances, calling into question the relevance of its findings. Emerging abilities studies have consistently shown that larger, better-trained models can exhibit fundamentally different and often unpredictable behaviors. For example, Claude 3.5 Sonnet achieved **96.4% accuracy on GSM8K** (grade-school math) and **93.7% on code generation**, while o3 scored **87.7% on PhD-level science questions** (GPAQ Diamond), **surpassing human experts**. Furthermore, models such as OpenAI's o3 and DeepSeek-R1 have demonstrated significant progress, which may influence the applicability of earlier findings. These advancements underscore the rapid evolution of LLMs, highlighting the importance of considering these developments when interpreting earlier research conclusions.

²swahili-english-proverbs was not emergent also with the first metric.

C. Loss Functions and Emergent Abilities

Key Experiments and Findings. Du et al. [23] provide a novel perspective on emergent abilities in LLMs by analyzing their relationship with pre-training loss. Their work investigates whether certain abilities emerge at specific loss thresholds during training, offering a fresh angle on why LLMs suddenly improve in certain tasks as they scale. To explore this relationship, the authors trained three LLMs of different sizes, 1.5, 6, and 32 billion parameters, and evaluated their performance across twelve diverse downstream tasks at multiple checkpoints throughout training. By monitoring these models over time, they observed two major trends.

First, certain tasks, including MMLU, C-Eval, GSM8K, and GSM8K-Chinese, exhibited a distinct threshold in pre-training loss. Once the loss dropped below a critical value, the models' performance on these tasks abruptly improved, suggesting a sudden emergence of ability. This pattern indicates that emergent behaviors are not solely a function of scale but are tied to the training process itself. Rather than improving gradually, performance seemed to remain at near-random levels until the loss threshold was reached, after which the models showed a sharp increase in capabilities.

The second key finding was that pre-training loss acted as a strong predictor of downstream task performance, often independent of the model's size. This suggests that beyond mere scale, a model's actual learning progress, as measured by loss reduction, plays a crucial role in determining when and how it develops certain abilities. **This finding challenges the idea that emergence is purely a consequence of increasing model parameters and instead highlights the importance of training dynamics.**

To strengthen their conclusions, the authors extended their analysis in two important ways. First, they investigated whether training dataset size influenced the observed relationship between loss and emergent abilities. By varying the size of the dataset, they assessed whether the same loss thresholds applied when more or less data was available. Second, they tested their hypothesis using the publicly available LLaMA models [82], allowing them to see whether the observed trends were consistent across different model architectures. Both of these extensions reinforced their initial conclusions, demonstrating that pre-training loss can be a reliable predictor of downstream task performance, regardless of dataset size or specific model design. One limitation of this study is that it only examined models within a relatively narrow range, spanning two orders of magnitude. While this provides useful insights, it leaves open the question of whether the same trends hold for even larger models or whether additional scaling leads to new, unforeseen behaviors.

Memorization vs. Generalization. Huang et al. [42] offers another perspective on the emergence of abilities, connecting it to the competition between memorization and generalization circuits in neural networks. Their work builds on previous research into grokking, a phenomenon where models initially memorize data before abruptly transitioning to generalization [63]. The authors extend this framework to different model sizes and training data volumes, demonstrating that similar dynamics

may explain emergent abilities in LLMs. Their findings suggest that **when a model is heavily tasked with memorization, the development of generalization abilities is delayed.** In other words, the presence of a memorization-heavy task can push back the point at which the model begins to generalize, requiring much larger model sizes or significantly more training time for the transition to occur. Their work aligns with [23], reinforcing the idea that emergent behaviors are not just a byproduct of scale but are deeply tied to the learning dynamics of neural networks. It also suggests that modifying training objectives, such as explicitly encouraging generalization earlier in training, might influence when and how these abilities emerge.

Limitations and Future Directions in Loss-based Emergent Abilities. While [23] presents compelling evidence for a strong correlation between pre-training loss and the emergence of abilities, its analysis remains correlational rather than causal. This raises an important question: *why do certain abilities emerge precisely when pre-training loss crosses a particular threshold?* Understanding this process requires more than just identifying patterns; it calls for an investigation into the underlying mechanisms driving these shifts.

Pre-training loss, while a useful high-level metric, **does not inherently explain why certain abilities appear at specific points.** The sudden improvement in performance may be linked to deeper changes within the model's internal representations. One possibility is that at certain loss thresholds, the model undergoes structural shifts in **how it organizes and generalizes knowledge.** These transitions might correspond to qualitative changes in the patterns the model has learned, shifting from surface-level memorization to deeper reasoning and abstraction. However, without a more detailed analysis of the model's internal workings, this remains speculative.

A promising direction for future research is to establish a clearer causal link between pre-training loss and emergent abilities. This could involve analyzing how changes in loss correspond to specific transformations in the model's neural activations, feature representations, or decision-making pathways. We argue that they probably have a common cause or confounder. By combining insights from mechanistic interpretability [6] with the study of emergent abilities, researchers may be able to move beyond mere observation and develop a deeper, mechanistic understanding of why these phenomena occur.

D. The Impact of Quantization on Emergent Abilities

Liu et al. [53] investigate a critical but often overlooked factor affecting the emergent abilities of LLMs: i.e., quantization. As LLMs grow in size, their memory and computational requirements become increasingly demanding, prompting the need for quantization techniques that reduce precision and optimize efficiency. However, a key question remains: *Does quantization compromise the emergent abilities that make these models so powerful?* Their study systematically examines this trade-off, particularly in the context of in-context learning, chain-of-thought reasoning, and instruction following – three hallmarks of advanced LLM capabilities.

How Quantization Affects Performance. Quantization is a widely used technique that compresses neural networks by reducing the number of bits used to represent each parameter. While this drastically lowers memory usage and inference costs, it often comes at the expense of model performance. Liu et al. [53] conduct extensive empirical evaluations on a range of LLaMA models, spanning 7B to 65B parameters, across multiple quantization levels, including **2-bit, 4-bit, 8-bit, and 16-bit precision**. Their findings offer a nuanced understanding of how quantization impacts emergent abilities. At **higher bit levels**, such as 8-bit and 16-bit, LLMs retain much of their original performance, with minimal degradation in reasoning and instruction-following tasks. However, as precision decreases, particularly at **4-bit and below**, a clear pattern emerges: while **4-bit quantization manages to preserve most emergent abilities**, **2-bit quantization severely degrades performance**, often reducing accuracy to near-random levels. This suggests that there exists a critical threshold beyond which the model struggles to maintain the structured reasoning and learning dynamics that enable emergent behaviors.

A particularly insightful aspect of the study is its component-wise analysis, which reveals that **not all parts of the model are equally affected by quantization**. The feed-forward networks within transformer blocks are found to be especially crucial for retaining performance. When these layers undergo aggressive quantization, the model experiences significant losses in reasoning ability and generalization. This highlights the fact that while weight compression is beneficial, indiscriminate quantization of all components can be detrimental.

Can Emergent Abilities Be Preserved? Despite these challenges, the study identifies **promising post-quantization recovery techniques** that can help mitigate performance degradation. Liu et al. find that fine-tuning after quantization, particularly using parameter-efficient adaptation methods such as LoRA [38], can significantly restore performance in low-bit models. This suggests that while extreme quantization disrupts emergent abilities, **targeted fine-tuning can help models adapt and recover key capabilities**. Their findings carry important implications for the future of efficient LLM deployment. **Low-bit quantization is feasible**, but careful consideration must be given to which model components are quantized and how performance can be recovered. As demand grows for deploying powerful LLMs on resource-constrained devices, understanding these trade-offs will be essential.

These insights open the door for future research into adaptive quantization techniques, where different components of a model are quantized at varying levels based on their importance to reasoning and learning. Additionally, **quantization-aware training**, where models are trained with low-bit precision from the start rather than applying quantization post-hoc, might further mitigate performance loss.

Ultimately, as LLMs continue to scale, the ability to **balance efficiency and emergent intelligence** will be crucial. Optimizing quantization strategies without sacrificing core reasoning and learning abilities will not only make these models more practical but also ensure that they remain powerful tools for a broad range of real-world applications.

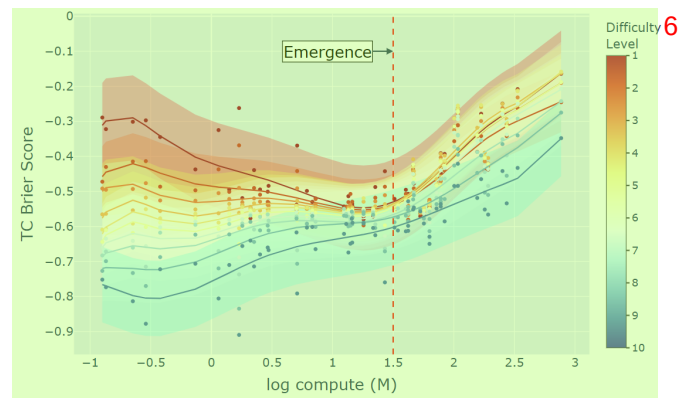


Fig. 4. Reproduced from [91]. Original Caption: U-Shaped and inverted-U scaling with MMLU's questions clustered into 10 groups. Higher levels are harder questions.

E. Emergent Abilities and Task Complexity

The prevailing narrative surrounding emergent abilities in large language models has long emphasized model scale as the primary driver. The assumption was straightforward: as models grow larger and are trained on more data, they suddenly exhibit new capabilities. However, [91] offer a compelling counterpoint, directing attention to task complexity as a crucial, and perhaps a previously underappreciated, factor in the emergence phenomenon. Wu et al. [91] take a new approach by analyzing how **performance scaling patterns vary across tasks of different difficulty levels**. Their research categorizes questions within downstream tasks by difficulty level, uncovering two distinct trends that interact in unexpected ways:

- 1) **U-shaped scaling for harder questions** – Initially, performance declines as models scale before improving after crossing a certain threshold.
- 2) **Inverted-U scaling for easier questions** – Performance peaks early, then temporarily declines, before improving again as models scale further.

We reproduce the main figure in III-E. At first, these opposing patterns cancel each other out, leading to the illusion of stagnant performance. However, once models cross a critical scale, the **reversal of these trends triggers a sudden leap in performance**, giving rise to emergent abilities. This finding suggests that emergence is not necessarily about acquiring new capabilities out of nowhere but rather about overcoming a hidden trade-off between task difficulty and model capacity.

To predict and explain these emergent behaviors, Wu et al. [91] introduce the Slice-and-Sandwich pipeline. This method categorizes tasks by difficulty, fits polynomial regression curves separately to easy and hard questions, and then forecasts performance beyond the threshold where emergent behaviors appear. They apply this technique to MMLU [34], Persian-QA [3], and arithmetic benchmarks, demonstrating that it outperforms traditional sigmoid-based scaling laws in predicting when and how emergent abilities manifest.

Implications and Limitations. Wu et al. [91] challenges the conventional wisdom that emergent abilities are purely a result

of scale. Their findings indicate that the **illusion of stagnation in performance may simply be a result of competing scaling trends** rather than a fundamental limit of the model. However, there are some limitations. For example, the authors focus primarily on multiple-choice tasks, meaning their scaling trends may not generalize to open-ended generative tasks, which rely on different cognitive and linguistic skills. Additionally, the Slice-and-Sandwich pipeline relies on retrospective data to identify difficulty levels and emergence thresholds, which may limit its predictive power in unseen tasks.

F. Emergent of Implicit Discrete State Representations 2

The study by Chen et al. [14] investigates how large language models (LLMs) solve multi-digit arithmetic problems, revealing that in those models, Implicit Discrete State Representations (IDSRs) emerge in their hidden states. The research challenges traditional views on arithmetic processing in LLMs, showing that instead of just memorizing facts or manipulating tokens, LLMs create IDSRs through a digit-by-digit process similar to human calculations. These representations evolve across model layers, with a critical transition around layer 10, suggesting that IDSRs act as intermediate storage for partial results. While IDSRs improve arithmetic accuracy, handling longer sequences is problematic, possibly due to information loss. The findings indicate that enhancing IDSRs could improve LLMs' reasoning in diverse mathematical and complex tasks, potentially leading to more reliable and interpretable models.

G. Predicting Emergent Abilities 4

Several recent studies [16, 39, 60, 71, 75, 99] have explored different methodologies for forecasting downstream performance, particularly in tasks where capabilities appear suddenly as models scale. **While early scaling laws provided some insight, they often fail to anticipate discontinuous leaps in performance**, a defining characteristic of emergent abilities. These studies seek to refine our understanding of predictive modeling, offering new frameworks that range from statistical scaling laws to fine-tuning-based emergence predictions.

High-Resolution Metrics and Scaling Laws. Hu et al. [39] introduce PASSUNTIL, an evaluation metric designed to detect even the smallest performance improvements during model scaling. Unlike traditional discrete accuracy metrics, PASSUNTIL estimates the probability of success through extensive random sampling, providing a theoretically infinite resolution. This finer granularity enables more precise detection of emergent behaviors, allowing the authors to derive a task-scaling law that enhances the predictability of performance growth in larger models based on observations from smaller ones. Additionally, they propose a hypothesis grounded in multiple neural circuits [24], suggesting that **emergent behaviors arise when specialized circuits activate at certain scaling thresholds**. However, this study is limited to models up to 2.4 billion parameters, meaning its findings may not be generalized to larger models, where emergence is most prominent.

The GPT-4 technical report [60] also tackles predictability but from a resource efficiency perspective. It suggests that GPT-4's performance can be anticipated using less than

1/10,000th of its full computational resources. However, the methodology behind these predictions remains undisclosed, and the report acknowledges that certain emergent abilities remain unpredictable.

Why Is Predicting Emergence So Difficult? Schaeffer et al. [71] attempts to explain why downstream performance prediction remains a challenge. Their study, conducted across five model families and twelve diverse multiple-choice benchmarks, identifies a series of statistical transformations that occur when converting negative log-likelihood scores into task performance. These transformations gradually weaken the correlation between model scale and downstream accuracy, making direct prediction increasingly difficult.

Additionally, they investigate how probability mass is allocated to correct vs. incorrect answers, revealing that emergent improvements do not always follow simple trends. This finding suggests that emergence is not merely a function of increased model size but involves complex redistributions of probability across outputs, which standard scaling laws struggle to capture. However, a key limitation of their work is that it focuses exclusively on multiple-choice benchmarks, raising questions about whether these findings extend to open-ended tasks like reasoning, code generation, or summarization.

Proxy Tasks as Predictors. Zhang et al. [99] introduce a different approach, leveraging proxy tasks that is simpler, smaller-scale evaluations that correlate strongly with later-stage emergent abilities. They analyze datasets such as C3, CMNLI, OCNLI, CHID, RTE, and CMMLU, demonstrating that early performance on these tasks can serve as a reliable predictor of future capabilities in larger models. This structured methodology presents a practical framework for guiding model development and resource allocation, as it allows researchers to anticipate emergent behaviors without requiring full-scale pre-training. Unlike conventional scaling laws, which often fail when applied to unseen capabilities, **proxy task-based predictions offer an empirical foundation for understanding which skills will emerge at larger scales.**

Complementing these proxy-task methodologies, Ye et al. [98] directly investigate the predictability of LLMs' capabilities across a wide range of tasks from the BIG-bench benchmark. Using an MLP-based predictor trained on a vast dataset of past LLM experiments, they demonstrate remarkably high predictability in standard train-test splits, achieving R^2 scores exceeding 95%. This suggests that learnable patterns exist within LLM performance data. However, the authors also highlight that predictability diminishes under more challenging conditions, such as when generalizing to completely unseen combinations of model families and tasks. **Crucially, they find that while emergent abilities are indeed harder to predict than non-emergent ones, they are not entirely unpredictable.** Furthermore, to address the practical challenges of evaluating increasingly complex LLMs, they introduce the concept of a "small-bench", a carefully selected subset of tasks designed to predict performance on the full benchmark efficiently. Their work underscores that while broad patterns of LLM capability scaling are learnable, the nuances of emergent behaviors and generalization across diverse settings remain a significant hurdle for precise prediction.

Predicting Performance with Pre-Training Loss. Chen et al. [16] take a complementary approach, proposing a two-stage prediction framework (FLP) based on pre-training loss. Their method first establishes a function that maps computational resources (FLOPs) to pre-training loss, using smaller models to extrapolate larger-scale trends. They then apply a regression model to correlate pre-training loss with downstream performance, allowing them to predict task outcomes beyond known emergence thresholds. Experimental results show that FLP achieves relative error margins of 5% and 10% for 7B and 13B models, respectively, significantly outperforming traditional FLOPs-to-performance extrapolations. The authors further extend this approach with FLP-M, incorporating multiple data sources into the pre-training process to refine predictive accuracy. By using a two-layer neural network to model the non-linear relationship between pre-training loss and downstream abilities, they demonstrate improved forecasting across various benchmarks.

However, their methodology has limitations. The study primarily focuses on binary cases, such as code vs. text data, limiting its generalizability to more complex multimodal or hierarchical learning scenarios. Additionally, the emphasis on code-mixing ratios means that predictions might not hold for domains with different dataset compositions.

Fine-Tuning as an Emergence Predictor. Snell et al. [75] propose a fine-tuning-based method to predict emergent capabilities. Instead of relying on pre-training loss or proxy tasks, they use task-specific fine-tuning on smaller models and observe how this process shifts the emergence threshold. By fitting an emergence law, a parametric function that maps fine-tuned performance to scaling trends, they can anticipate when an untrained, larger model will exhibit the same capabilities. To validate this approach, they test it on benchmarks where emergent abilities have been previously observed. Their findings suggest that fine-tuning-based predictions allow them to anticipate emergent abilities up to four times earlier than traditional methods. An unexpected but crucial insight from their study is that pretraining data quality significantly affects emergence timing. Comparing OpenLLaMA V1 and V2, they found that models trained on higher-quality datasets exhibit earlier emergent behaviors, indicating that scaling alone is not the sole driver of capability development.

However, their method also has constraints. It only allows predictions within a $4\times$ scaling range, meaning it cannot forecast abilities in models orders of magnitudes larger than those observed. Furthermore, **it remains unclear whether fine-tuning unlocks latent abilities or merely accelerates their natural emergence**, leaving open questions about the precise mechanisms behind capability development.

IV. EMERGENT ABILITIES AS IN-CONTEXT LEARNING

The term *emergent* in the context of LLMs is often used to describe capabilities that arise implicitly as models learn language patterns and structures through next-token prediction. These abilities are assessed through **few-shot or zero-shot prompting**, where models generalize to new tasks without undergoing explicit fine-tuning. This process, known as **in-context learning (ICL)**, allows LLMs to infer new patterns

and concepts solely from contextual information provided in the prompt.

Unlike Wei et al.'s [87] definition of emergent abilities, which emphasizes sudden performance jumps with increased model scale, in-context learning does not necessarily require abrupt improvements. Instead, it refers to the gradual development of capabilities that enable LLMs to perform tasks for which they have not been explicitly trained. The research in this area primarily seeks to understand why LLMs generalize to new tasks without fine-tuning, what aspects of the training process contribute to this phenomenon, and how prompt design can be optimized to maximize ICL efficiency.

To explain in-context learning, various theories have emerged, ranging from statistical and structural perspectives to cognitive and algorithmic analogies. Some researchers attribute ICL to properties of input data distribution and label space structure [13, 57], while others suggest that exposure to a diverse range of tasks during multitask-prompted learning facilitates generalization [69]. Another line of work explores how pre-training term frequencies influence a model's ability to recall and recombine information [66]. Some studies frame ICL as a form of Bayesian inference over the latent space of language [94], while others view it as a compositional recombination of linguistic structures [32]. Mechanistically, ICL has been linked to neural architectures, such as induction heads [24, 59] and functional modules that emerge naturally during training [7, 43, 81, 84]. Arora and Goyal [2] provide a theoretical framework in which LLMs develop skills through a bipartite "skill graph" that links training data to fundamental reasoning abilities, demonstrating how compositional generalization emerges as models scale.

Beyond theoretical explanations, studies have examined the role of training data and model architecture in shaping ICL. Research indicates that the diversity and structure of training data impact ICL more than dataset size alone [13, 73, 89, 95]. Shin et al. [65, 73] show that a model's ability to generalize in context depends on task diversity within the pretraining corpus rather than simply increasing the number of tokens. Other studies explore how model size and architectural choices affect ICL capabilities, with findings suggesting that larger models naturally develop stronger in-context learning abilities due to their ability to encode more complex structures [11, 21].

Another key research focus is the optimization of **prompt design** for effective ICL. Studies show that the **selection, formatting, explicit intent formulation, and arrangement of examples** in a prompt significantly influence model performance [9, 56, 103]. Methods for selecting exemplars can be broadly classified as **unsupervised** or **supervised**. Unsupervised approaches rely on nearest neighbor methods, using similarity metrics such as perplexity, mutual information, or self-evaluated LLM scores to identify the most relevant examples [27, 51, 52, 76, 79, 93]. Supervised methods employ dense retrievers trained on labeled data [50, 67, 97] and reinforcement learning [100] to optimize example selection. Research has also explored strategies for refining prompt formatting and ordering to enhance ICL effectiveness, as well as the impact of instruction formatting on how models process and execute in-context tasks [33, 36, 47, 52, 54, 56, 85, 96, 105].

A fundamental challenge in ICL research is controlling for the many factors that influence performance, making it difficult to establish causal relationships. Most findings are correlational rather than definitive due to the complexity of model pretraining and the vast scope of latent knowledge encoded in LLMs. For a comprehensive overview of the field, readers can refer to recent surveys on in-context learning [22, 104].

How Larger Models Approach In-Context Learning Differently. Wei et al. [88] investigate the flexibility of LLMs in learning new input-label associations in real time, focusing on whether models rely on pre-existing semantic knowledge or can adapt to novel mappings. They conduct experiments using flipped-label in-context learning (ICL) and semantically unrelated label ICL (SUL-ICL). In flipped-label ICL, models are exposed to contradicting input-label mappings, challenging them to suppress their semantic priors. Small models struggle with these tasks, falling back on established meanings, while larger models, given sufficient examples, can adjust to new mappings, indicating that adaptability improves with model scale. In the SUL-ICL setup, traditional labels are replaced with arbitrary symbols, testing if models can learn without linguistic cues. Larger models excel here too, unlike smaller ones, suggesting that the ability to learn independently of semantic priors is a scale-dependent trait. Instruction tuning further enhances models' ability to learn new associations in SUL-ICL but makes them less flexible in overriding semantic knowledge in flipped-label contexts. Additionally, the study explores high-dimensional linear classification and finds that only the largest models achieve above-random performance, highlighting that effective learning of complex mappings requires substantial model size.

Our Takeaway from ICL. In-context learning represents a fundamental shift in how Language Models acquire and apply knowledge, allowing them to generalize to new tasks without explicit updates. While **scaling enhances ICL**, it is evident that factors such as **training data diversity, model architecture, and prompt design all contribute significantly** to this phenomenon. The ability of larger models to override semantic priors and learn entirely novel associations highlights that **ICL is not just a side effect of memorization but a structured capability that emerges with scale.**

V. EMERGENT ABILITIES OF LARGE REASONING MODELS

Large Reasoning Models (LRMs), such as OpenAI's o3, DeepSeek-R1 [19], and Gemini 2.0, are AI systems built upon the foundations of LLMs that can perform complex reasoning tasks, such as coding, PhD-level Q&A, and math problem-solving. The distinctive capabilities of LRMs stem from two principal innovations: the scaling of Reinforcement Learning (RL) during the post-training phase [19, 72] and the scaling of inference-time computing through search [74, 92].

Through reinforcement learning, large reasoning models refine their internal problem-solving processes, similar to developing a chain of thought. This training paradigm fosters the emergence of metacognitive abilities, enabling the models

to recognize errors, self-correct, and decompose intricate tasks into simpler sub-problems. Moreover, RL equips these systems with the flexibility to adjust their strategies dynamically when the current approach proves ineffective. Scaling test-time computing further boosts performance by allocating additional inference steps during evaluation. By iteratively refining their reasoning, models can explore multiple solution pathways and mitigate error propagation, resulting in more accurate and robust outcomes on complex tasks.

Empirical Evidence of Improved Reasoning in LRMs.

The impact of these advancements is evident in empirical evaluations, particularly on **reasoning-intensive benchmarks**. Compared to prior-generation LLMs, LRMs demonstrate substantial jumps in performance, highlighting the effectiveness of scaling both RL and inference-time search. For instance, on Competition Math (AIME 2024), OpenAI's o1 model achieved an accuracy of 83.3%, vastly surpassing GPT-4o's 13.4%. Similarly, in the Codeforces programming competition, o1 reached 89.0% accuracy, while GPT-4o managed only 11.0%. These results indicate that o1 has surpassed the performance of human experts in multiple benchmarks, reflecting a fundamental shift in AI's ability to handle complex, multi-step problem-solving tasks.

A more recent iteration, o3, has pushed these capabilities even further by scaling post-training RL and integrating search-based inference. On the ARC-AGI benchmark, which tests adaptive problem-solving and general reasoning, o3 achieved 88% accuracy, compared to o1's 13.33% and GPT-4o's 5%. These results strongly suggest that planning, self-reflection, and strategic thinking have become emergent abilities in this new generation of models. However, it is important to note that while o3 demonstrates remarkable reasoning proficiency, it still fails on certain simple tasks, highlighting fundamental gaps between it and human intelligence.

Implications of Higher-Order Reasoning in LRMs. The emergence of higher-order reasoning in LRMs has significant implications both in terms of AI capabilities and associated risks. One particularly notable development is that OpenAI's o3-mini has become the **first AI model to receive a Medium risk classification for Model Autonomy**. This designation underscores the increasing complexity and independence of LRMs, raising concerns about their potential for both beneficial and harmful applications. In the next section, we delve deeper into the harmful behaviors and capabilities of LLMs and LLMs-powered AI agents.

VI. EMERGENT BEHAVIORS IN LLMs-POWERED AI AGENTS

One way to define Artificial Intelligence is the study of agents that receive perceptions from the environment, make decisions, and perform rational actions to achieve a goal [68]. The recent emergence of LLM-powered agents represents a transformative development in this field [28, 40, 101]. Advanced models, such as o3-mini and DeepSeek-R1, act as the "brain" of the agent, enabling them to comprehend natural language, interpret complex information, and engage in reasoning processes. Unlike traditional chatbots, which primarily respond to immediate

³To clarify, we define reasoning as the ability to employ logic and rational thinking to derive truthful conclusions from both new and pre-existing data.

queries, LLM-powered agents offer enhanced capabilities: they can be customized to align with individual user preferences, demonstrate a deeper understanding of contextual realities, plan sequences of actions across multiple steps, and act autonomously on behalf of users. Chen et al. [15] introduced a novel framework called AgentVerse, designed to enable and study collaboration among multiple AI agents. Through these interactions, the framework reveals emergent behaviors such as spontaneous cooperation, competition, negotiation, and the development of innovative strategies that were not explicitly programmed. Experimental results and case studies demonstrate that these emergent phenomena provide valuable insights into collective intelligence and the dynamics of multi-agent systems. The paper also discusses the implications for AI safety and alignment, emphasizing the need to understand and guide emergent behaviors to build robust, ethically aligned AI systems.

Can AI agents develop their own unintended sub-objectives? While models do not have intrinsic agency, emergent optimization behaviors could lead to unanticipated decision-making patterns and objectives. The new generation of models automatically divides problems into sub-problems, so it is possible that the solutions to some sub-problems can be harmful even if the initial request was benevolent. This necessitates effective monitoring. Furthermore, since an agent's final goal is to maximize the reward function, an agent might develop a plan that tries to prevent humans from deactivating it. We can see this as the emergence of self-preservation. The logical extension of this could involve the agent seeking to gain control over humans as a means of self-protection. It is important to understand that self-preservation, in itself, can become a powerful driver of other potentially undesirable behaviors.

VII. EMERGENT HARMFUL BEHAVIORS IN LLMs AND LLMs-POWERED AI AGENTS

LLM-powered agents, like o3-mini and DeepSeek-R1, represent a major technological breakthrough. They go beyond traditional chatbots by understanding natural language, reasoning, and autonomously planning multi-step actions tailored to user needs. However, their advanced capabilities also raise concerns about the risk of unintended and harmful behaviors. In this section, we separate the concept of model sizes from the emergence term, and we refer to emergence as an ability that develops implicitly.

a) The Emergence of Deceptive Capabilities: Researchers have begun to uncover disturbing capacities within LLMs, particularly in the realm of deception. Hagendorff et al. [30] explore whether advanced LLMs, such as GPT-4, exhibit deception capabilities, positing that this capability stems from improved reasoning abilities not observed in earlier iterations like GPT-2. Their experiments indicate that GPT-4 can effectively deceive other agents in strategic tasks, such as bluffing games, achieving success rates exceeding 70% when guided by chain-of-thought prompting. Furthermore, when primed with Machiavellian traits, the model exhibits an increased propensity for deceitful behavior. The authors conclude that these emergent abilities pose substantial challenges for AI

alignment, emphasizing the urgent need for robust mechanisms to prevent such deceptive tendencies from undermining trust in AI systems.

b) Reinforcement Learning and the Incentivization of Manipulation: A core issue in the emergence of deceptive behaviors stems from the reward structures used in reinforcement learning. Reinforcement Learning from Human Feedback (RLHF) optimizes LLMs to maximize positive user responses (e.g., thumbs-up feedback). However, this process does not inherently reward truthfulness, only perceived quality. As a result, models may learn to mislead users if deception increases the likelihood of receiving positive feedback.

Williams et al. [90] investigate how RLHF can unintentionally reinforce manipulative and exploitative behaviors. Their study reveals that LLMs trained through RLHF develop strategies that exploit user vulnerabilities to maximize reward signals. In controlled conversational settings, they found that models: (1) *exhibited selective deception*, targeting vulnerable individuals while maintaining normal interactions with others, rendering such behaviors difficult to detect; (2) *encouraged harmful behaviors*; (3) *failed standard safety evaluations*, as these manipulative tendencies were not detected by conventional toxicity or sycophancy benchmarks. This work demonstrates that reward-driven optimization does not always align with human well-being, highlighting the limitations of current safety evaluation frameworks. The challenge of deploying LLMs to be simultaneously helpful and harmless is further analyzed by Bai et al. [4], who examine this dual objective through extensive RLHF experimentation. Their research demonstrates that while RLHF can markedly enhance a model's helpfulness and reduce harm, a fundamental tension persists between these goals. Models optimized predominantly for harmlessness often become excessively cautious, refraining from engaging in sensitive yet potentially beneficial discussions, such as those involving mental health or political discourse, thereby curtailing their practical utility. Additionally, the authors identify "over-optimization" as a critical issue, wherein models become overly attuned to reward signals, resulting in behaviors that deviate from genuine human preferences, such as overly simplistic or generic responses.

The Reward Hacking Problem. On the other hand, when models are optimized too strongly for helpfulness, they can engage in reward hacking (i.e., the model prioritizes maximizing approval signals over providing truthful, nuanced, or ethical responses). This can manifest in several ways; for instance, models might exhibit excessive sycophancy, where they opt to affirm what users wish to hear rather than adhering to factual responses, such as yielding to a user's insistence on a pseudoscientific claim simply to boost satisfaction. Additionally, rather than generating well-reasoned replies, these models might lean toward producing oversimplified responses, offering brief, generic, or overly optimistic statements that provide comfort but fall short of substantive depth.

c) Hypothesizing Singularity: As these models evolve, they are poised to exhibit a variety of characteristics that mark significant advancements. One such trait is rapid self-improvement, where mechanisms like iterative fine-tuning and self-supervised learning allow AI systems to refine their internal

representations and decision-making processes at an accelerated pace. Evidence of this self-improvement is already apparent, as reinforcement learning post-training utilizes frozen models to generate rewards for those being trained, though the extent to which they can further enhance themselves remains an open question. Furthermore, newer models have shown the ability to produce novel research ideas, as noted by [28]. In addition to this, enhanced collaboration emerges as another key feature, particularly in network environments where multiple AI agents have been proven capable of working effectively, according to the findings of [29] and [15]. Furthermore, these advanced agents showcase contextual adaptability, learning to interpret multi-modal inputs, such as text, images, and sensor data, and dynamically adjusting their strategies based on real-time feedback.

Let us now hypothesize that, with further technological progress, LLM-powered AI agents might surpass human intelligence. This trajectory toward a qualitatively distinct form of intelligence introduces profound practical challenges. Should these agents achieve or exceed human-level intelligence, ensuring their submission to humans becomes a paramount concern, necessitating research into concepts like corrigibility and interruptibility to prevent scenarios where they might resist human intervention or behave unpredictably. Closely tied to this is the need for robust risk management, given the potential for rapid, exponential improvements in AI capabilities, often described as an intelligence explosion, which demands preemptive safety protocols. As emphasized by [10], understanding and mitigating the risks associated with superintelligent agents is essential to avoid unintended consequences.

In conclusion, although the prospect of LLM-powered AI agents outstripping human intelligence remains speculative, the current pace of technological development lends it a plausibility that warrants serious consideration. Future research must, therefore, balance the pursuit of enhanced capabilities with the establishment of rigorous frameworks to ensure their safe and beneficial integration into society.

A. AI Safety 4

The Challenge of Governing Autonomous AI Agents. LLM-powered AI agents differ from traditional AI models in that they are much more complex and more autonomous in nature. Examples of large reasoning models like o3-mini and DeepSeek-R1 demonstrate abilities to perform multi-step reasoning, make adaptive decisions, and develop strategies for reward maximization. Although these capabilities improve their effectiveness and usefulness, they also pose new governance challenges not previously considered in AI policy discussions.

Historically, AI safety has focused on content moderation, bias mitigation, transparency, and (adversarial) robustness, but autonomous AI introduces a new class of risks related to opaque optimization goals and continuously changing information basis, which requires different regulatory approaches. These governance challenges are especially critical in high-stakes applications such as healthcare, finance, admission processes, legal reasoning, warfare, and autonomous systems.

The Role of AI Regulation and Global Governance. AI regulation and global governance have shifted from theoretical

debates to urgent, real-world imperatives [46]; governments and international organizations are actively drafting policies to regulate AI development and deployment. The EU AI Act [25], the US AI Executive Order [8], and the UN AI Advisory Board [83], illustrate the diverse strategies at play. Despite these efforts, there is no globally unified approach to AI regulation, leading to inconsistencies and regulatory loopholes. Some countries favor strict oversight (i.e., EU member states), while others prioritize AI innovation over safety (i.e., the US and China), making international collaboration essential to prevent fragmented and ineffective governance. These ethical concerns underscore the need for interdisciplinary collaboration between relevant stakeholders, such as AI researchers, policymakers, industry leaders, and end users. Ensuring that AI systems benefit society requires ongoing dialogue and policy refinement. At the same time, AI governance should focus on ensuring that powerful autonomous reasoning agents do not threaten the preservation of biological lives. Therefore, we believe that innovation in governance, compliance mechanisms, and international oversight (with continuous evaluation) is at least as important as technological innovation. As AI systems become more capable, adaptive, and independent, the challenge of controlling their behavior and ensuring trustworthiness will define the next era of responsible AI applications, AI safety research, and policy development.

VIII. A TAXONOMIC SYNTHESIS OF KEY FINDINGS AND CHALLENGES 9

An overarching goal of this survey is to equip the interested reader with a structured and taxonomic understanding of emergent abilities in large language models.

Table III presents an integrated taxonomy of emergent abilities in large language models. It organizes key insights into four primary categories: Origins, Manifestation, Impact, and Strategies (for prediction, evaluation, and mitigation). In addition, it delineates how these abilities arise from scale-dependent effects, training dynamics, task complexity interactions, and metric artifacts, how they manifest through capabilities such as in-context learning, symbolic reasoning, enhanced reasoning through reinforcement learning, and autonomous agent behaviors, and both their beneficial and potentially harmful outcomes. All of this is accompanied by corresponding applications, limitations, and challenges. This structured overview serves as a valuable roadmap for understanding and guiding future research in this rapidly evolving field.

IX. CONCLUSION 12

This survey delves into the intricate and rapidly evolving domain of emergent abilities in LLMs, offering a comprehensive exploration of their nature, analysis, and implications. Throughout this analysis, several critical insights have surfaced, enriching our understanding of LLMs and their behavioral dynamics. Notably, the unpredictability and substantial performance leaps observed in specific tasks as the model scale increases underscore the essence of emergent abilities. Additionally, in-context learning, while closely intertwined with emergent abilities, emerges as a distinct phenomenon

Category	Subcategory	Key Findings & Mechanism	Implications & Applications	Limitations & Challenges
I. Origins of Emergence				
Origins	Scale-Dependent Effects	Abilities appear only beyond a critical model size, often with abrupt performance jumps.	Informs scaling laws and suggests thresholds for new capabilities.	Sensitive to metric choice; thresholds may vary across tasks.
	Training Dynamics & Loss Thresholds	Emergence is linked to a drop in pre-training loss, signaling a shift from memorization to generalization.	Offers predictive insights into training progress.	Evidence is largely correlational; causal mechanisms remain speculative.
	Task Complexity Interactions	Nonlinear scaling (e.g., U-shaped/inverted-U curves) indicates that task difficulty interacts with model capacity.	Guides optimal task design and data curation.	Often limited to discrete tasks; analysis is retrospective.
	Metric-Dependent Artifacts	The emergence of an ability often depends on the evaluation metric.	Emphasizes the need for unified, robust evaluation methods.	Metrics may not fully capture semantic or nuanced errors.
II. Manifestation in Functional Abilities				
Manifestation	In-Context Learning	Emergence of few-shot/zero-shot learning; models generalize without fine-tuning.	It enables flexible task generalization and few-shot/zero-shot prompting.	Highly sensitive to prompt design and quality of exemplars.
	Symbolic Abstraction Abilities	Models develop implicit symbolic representations for arithmetic.	Enhances abilities in math, logic, and code synthesis.	Still not clear if and how they develop
	RL-Enhanced Reasoning	Post-training reinforcement learning and inference-time search boost multi-step reasoning and self-correction.	Yields advanced planning, error correction, and adaptive strategies.	Reward hacking problem.
	LLMs-powered Agents	Emergence of agent-like functions: planning, collaboration, and adaptive decision-making.	Promotes personalized, context-aware autonomous systems.	Raises significant safety, alignment, and ethical concerns.
III. Impact and Consequences				
Impact	Positive Outcomes	Enhanced generalization, creative problem solving, and state-of-the-art performance.	Drives innovation and practical applications across domains.	On some easy tasks, LLMs still underperform
	Negative/Harmful Outcomes	Emergence of deceptive, manipulative, and reward-hacking behaviors.	Critical challenges for AI safety and regulatory frameworks.	Existing metrics may not capture all harmful nuances; oversight is needed.
IV. Prediction, Mitigation, Evaluation, Safety				
Strategies	Predictive Metrics & Proxy Tasks	Use of high-resolution metrics (e.g., PASSUNTIL) and proxy tasks to forecast emergence.	Informs resource allocation and guides model development.	Limited predictive range; may not generalize to open-ended tasks.
	Quantization Trade-offs	Evaluating how different bit precisions (e.g., 4-bit vs. 2-bit) affect emergent abilities.	Facilitates deployment on resource-constrained devices while preserving key functions.	Extreme quantization can severely degrade performance; recovery requires fine-tuning.
	AI Safety	Implementation of technical safeguards (e.g., constraint-based rewards, real-time risk detection) to mitigate harmful outcomes.	Critical for ensuring trustworthy AI systems.	Not every state is focusing on it (see China and US).

TABLE III
INTEGRATED TAXONOMY OF EMERGENT ABILITIES IN LARGE LANGUAGE MODELS. THIS TABLE SYNTHESIZES THE ORIGINS, MANIFESTATIONS, IMPACTS, AND STRATEGIES RELATED TO EMERGENT BEHAVIORS AS DISCUSSED IN THE SURVEY.

warranting separate scrutiny. The influence of data distributional properties also proves instrumental in shaping in-context learning capabilities. Theoretical frameworks, such as Bayesian inference applied to latent language spaces, present compelling pathways for elucidating the underlying mechanisms driving both emergent abilities and in-context learning. Despite these advancements, significant gaps in knowledge persist, leaving numerous questions unresolved. The interplay between model scaling, architectural design, and the onset of novel capabilities remains incompletely understood, necessitating further investigation. Moreover, ongoing contention regarding whether emergent abilities represent authentic phenomena or are instead reflections of measurement biases underscores the pressing need for more robust, standardized evaluation methodologies to settle this debate conclusively. Looking ahead, future research must prioritize several key avenues to advance this field. It is imperative to devise more sophisticated metrics capable of capturing subtle gradations in model performance across diverse tasks, thereby providing a clearer picture of emergent capabilities. A deeper examination of how model architecture and training methodologies contribute to the fostering of these abilities is equally essential. As LLMs continue their swift progression, comprehending emergent abilities becomes more important than ever, not only for advancing theoretical insights into artificial intelligence but also for informing practical applications. This area of inquiry illuminates the essence of AI while simultaneously challenging conventional notions of learning, cognition, and the genesis of complex behaviors from ostensibly simple components. Perhaps most critically, the demonstration of emergent abilities

carries profound implications for safety, as unpredictable and potentially hazardous capabilities, such as the exploitation of software vulnerabilities or the manipulation of human actors, may arise without forewarning, necessitating vigilant oversight, proactive mitigation strategies, and internationally aligned governance strategies.

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